



Salman Khan

Date of birth: 16 Mar 2006 Phone number: (+92) 03137700673

Email address: salmankhan.pk211@gmail.com

Website: www.linkedin.com/in/salman-khan-bscs-student

Website: <https://github.com/SalmanKhanPK211/>

Website: <https://salman-khan-bscs-student.vercel.app/>

ABOUT ME

Motivated BS (Hons) Computer Science student with strong academic performance and practical experience at the intersection of AI/ML, IoT, embedded systems, and software development. Developed Water Angel, an ESP32-based **autonomous smart water-management and water-quality monitoring system** integrating real-time sensing, automated pump control, cloud synchronization, and a Progressive Web Application. Interested in research-oriented opportunities involving intelligent sensing, machine learning, embedded/edge systems, and data-driven solutions to real-world problems.

EDUCATION AND TRAINING

BS (Hons)

University of Swabi [1 Sep 2024 – Current]

City: Swabi | Country: Pakistan | Website: <https://uoswabi.edu.pk> | Field(s) of study: Computer Science | Final grade: CGPA 3.85/4.00 (through 4th semester) | currently in 5th semester

RESEARCH INTERESTS

Research interests:

Intelligent IoT systems, machine learning, embedded and edge AI, intelligent sensing, wireless and connected systems, and data-driven optimization.

WORK EXPERIENCE

Code Alpha – Remote

Data Analytics Intern

[20 Aug 2026 – Current]

Working on hands-on projects in web scraping, Exploratory Data Analysis (EDA), and data visualization using Python. Collected, cleaned, analyzed, and visualized real-world datasets using Pandas, BeautifulSoup, Matplotlib, and Seaborn, while identifying trends, patterns, anomalies, and data-quality issues and documenting projects using GitHub.

Digital Empowerment Network (DEN) – Remote

Web Development Intern

[15 Jul 2025 – 2 Sep 2025]

Developed responsive web applications using HTML, CSS, and JavaScript; built interactive projects including a To-Do List, Calculator, Quiz, Weather App, and Landing Page; practiced DOM manipulation, debugging, and Git/GitHub version control.

PROJECTS

[22 Oct 2025 – 2 Aug 2026]

Water Angel — Autonomous Smart Water Management & Water-Quality Monitoring System

- Designed and developed an ESP32-based system integrating water-level sensing, TDS water-quality monitoring, and automated pump control.
- Implemented dual-core FreeRTOS task isolation, reducing measured pump-control reaction time from **4,500 ms** (single-core) to **150 ms**.
- Developed an offline-first architecture with EEPROM-persisted thresholds, enabling autonomous operation during network outages.
- Built a Progressive Web Application (PWA) for remote monitoring, threshold configuration, and data visualization.

- Evaluated low-cost TDS sensing, achieving a **mean absolute error of 12.08 ppm** against a reference meter.
- Developed **XGBoost** model for short-term weekly household water-consumption forecasting.

Link: https://drive.google.com/drive/folders/1cKcpr-BxL1jLJFM3SU9hcQPv1ijtxSCU?usp=drive_link

SKILLS

Programming and Databases

Python / C++ / MySQL / VS Code / GitHub / Git

Data Science and ML

Machine Learning / NumPy / Matplotlib / Pandas / Power BI / Scikit-Learn / XGBoost

IoT and Embedded Systems

Arduino / ESP32 / Embedded Systems / FreeRTOS / Sensor Interfacing / Relay Control

LANGUAGE SKILLS

Mother tongue(s): Pashto

Other language(s): English

HONOURS AND AWARDS

[27 Jan 2026] University of Swabi

Awarded laptop under PM laptop scheme Awarded a laptop under the Prime Minister's Laptop Scheme based on academic merit.

[22 Jul 2026] University of Swabi

Honhar Scholarship Awarded a merit-based full tuition fee waiver in recognition of academic performance.

CERTIFICATIONS

[1 Oct 2025 – 7 Oct 2025]

IoT for Everyone

Coursera

[1 Sep 2026 – 4 Sep 2026]

Introduction to Data Science

Cisco Networking Academy

[8 Jun 2026 – 29 Jul 2026]

ACT AI (Awareness, Competency & Tools Training for Artificial Intelligence)

AI SkillBridge

CONFERENCES AND SEMINARS

[8 Jul 2026 – 8 Jul 2026] Ghulam Ishaq Khan Institute, Topi (Swabi)

Data to Decision: Mastering ML for Climate Solutions Participated in hands-on machine learning training using Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn, covering classification, regression, and ML workflows.

VOLUNTEERING

[6 Dec 2025 – 6 Dec 2025] Gohati College Swabi

Interview Panel Member Participated in applicant interviews and merit-based selection for Alkhidmat IT training programs.